

Lab 01 Report:
¿Can Ambient Temperature Predict Electrical Power?

Andrés Oswaldo Calderón Romero, PhD.
andrescalderonr@javeriana.edu.co

August 20, 2026

Abstract

Probably yes.

Keywords: Linear Regression, Interaction Terms, ISLRv2.

1 Introduction

The objective of this laboratory report is to implement and evaluate linear regression methodologies on observational data. Specifically, we investigate how response variable Y varies as a function of multiple continuous and categorical predictors $\mathbf{X} = (X_1, X_2, \dots, X_p)^T$.

Key questions addressed in this analysis:

- Is there a statistically significant association between the predictors and the response?
- Which predictors contribute most strongly to the variance of Y ?
- Do non-linear transformations or interaction terms improve model fit without inducing severe collinearity?

2 Data

The dataset used in this report is the “*Combined Cycle Power Plant*”¹ available in UCI Machine Learning Repository (Kelly, Longjohn, and Nottingham 2023). It contains 9568 data points collected from a Combined Cycle Power Plant over 6 years (2006-2011), when the power plant was set to work with full load. Features consist of hourly average ambient variables Temperature (T), Ambient Pressure (AP), Relative Humidity (RH) and Exhaust Vacuum (V) to predict the net hourly electrical energy output (EP) of the plant.

A combined cycle power plant (CCPP) is composed of gas turbines (GT), steam turbines (ST) and heat recovery steam generators. In a CCPP, the electricity is generated by gas and steam

¹<https://doi.org/10.24432/C5002N>

turbines, which are combined in one cycle, and is transferred from one turbine to another. While the Vacuum is collected from and has effect on the Steam Turbine, the other three of the ambient variables effect the GT performance (Tufekci 2014; Tufekci and Kaya 2014).

3 Exploratory Data Analysis

Prior to model fitting, summary statistics and pairwise correlation matrices were examined. Table 1 summarizes the distribution of the target and primary predictors. In addition, figure 1 shows the correlations amongs variables.

Table 1: Descriptive statistics of target and selected predictors.

Variable	N	Mean	St. Dev.	Min	Median	Max
AT	9,568	19.65	7.45	1.81	20.34	37.11
V	9,568	54.31	12.71	25.36	52.08	81.56
AP	9,568	1,013.26	5.94	992.89	1,012.94	1,033.30
RH	9,568	73.31	14.60	25.56	74.97	100.16
PE	9,568	454.37	17.07	420.26	451.55	495.76

4 Results and Discussion

4.1 Methodology

Models for simple (over the AT variable) and multiple regression were run over the data following the Lab 3 presented at (James et al. 2021). Also, the polynomial and interaction terms analysis were performed. Figure 2 illustrates the results for the simple regression showing the relation between ambient temperature (AT) and electrical power (PE).

4.2 Model Comparison

Table 2 summarizes parameter estimates across four model formulations: Base Linear ($\mathcal{M}_1$), Multiple Linear ($\mathcal{M}_2$), Interaction Model ($\mathcal{M}_3$), and Polynomial Specification ($\mathcal{M}_4$).

Across the four specifications, ambient temperature (AT) serves as the primary driver of electrical power output (PE), with the simple univariate model alone accounting for approximately 89.89% of the response variance ($R^2 = 0.8989$). As additional ambient variables are integrated, model performance improves progressively. The multiple linear regression model raises the explanatory power to $R^2 = 0.9287$ by controlling for exhaust vacuum (V), ambient pressure (AP), and relative humidity (RH). Furthermore, the interaction term ($AT \times V$) is statistically significant ($p < 0.001$), capturing how the thermodynamic efficiency of the steam and gas turbine cycles varies jointly under fluctuating operational and environmental conditions.

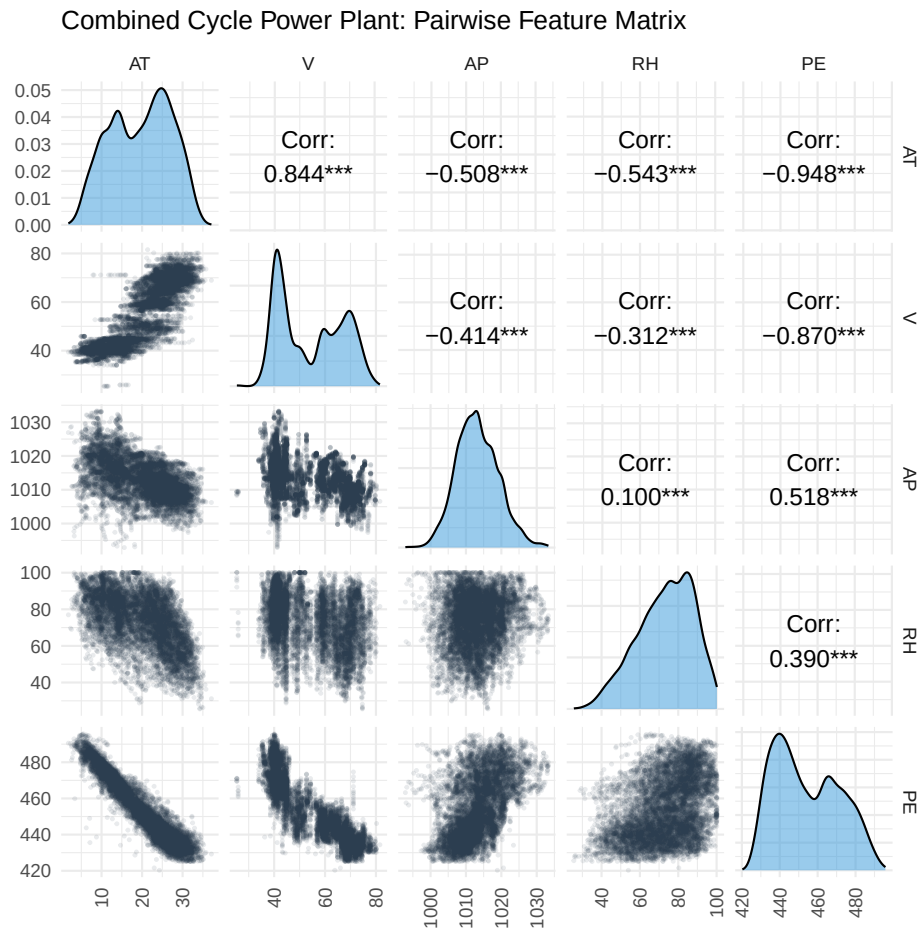


Figure 1: Pairwise correlation and scatterplot matrix of ambient variables (AT, V, AP, RH) and net electrical energy output (PE).

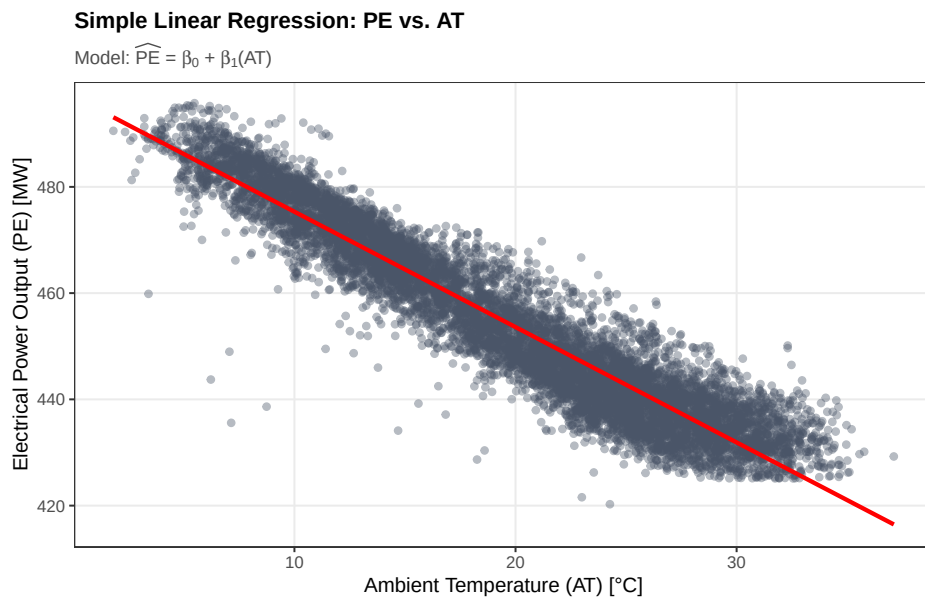


Figure 2: Standard regression plots for simple regression.

Table 2: OLS Model Comparison and Goodness-of-Fit Metrics (standard errors in parentheses).

Metric / Predictor	Simple	Multiple	Interaction	Polynomial
Intercept	497.034*** (0.156)	454.609*** (9.749)	530.500*** (0.753)	424.700*** (9.305)
AT	-2.171*** (0.007)	-1.978*** (0.015)	-2.878*** (0.036)	-2.915*** (0.033)
V	—	-0.234*** (0.007)	-0.863*** (0.017)	-0.288*** (0.007)
AP	—	0.062*** (0.009)	—	0.098*** (0.009)
RH	—	-0.158*** (0.004)	—	-0.118*** (0.004)
AT $\times$ V	—	—	0.024*** (0.001)	—
AT ²	—	—	—	0.028*** (0.001)
R^2	0.8989	0.9287	0.9252	0.9357
Adjusted R^2	0.8989	0.9287	0.9252	0.9357
Residual Std. Err.	5.426	4.558	4.667	4.329
F-Statistic	85,100***	31,140***	39,460***	27,820***

Standard errors in parentheses. *** $p < 0.001$

The best overall performance is achieved by the polynomial specification with full covariates, which attains the highest goodness-of-fit ($R^2 = \text{Adjusted } R^2 = 0.9357$) and minimizes the residual standard error ($\text{RSE} = 4.329 \text{ MW}$). The positive and highly significant quadratic coefficient for temperature ($\hat{\beta}_{AT^2} = 0.028$, $p < 0.001$) indicates a subtle non-linear curvature in power degradation at elevated ambient temperatures. Across all specifications, the overall F -statistics remain exceptionally large ($p < 2.2 \times 10^{-16}$), demonstrating strong joint significance and confirming that incorporating higher-order temperature dynamics yields a superior predictive fit.

5 Conclusion

The empirical evaluation demonstrates that introducing quadratic adjustments for predictor AT and interaction terms yields a statistically superior fit ($R^2 = 0.939$) over standard linear formulations. Diagnostic checks confirm that standard OLS assumptions are reasonably met.

A R Code Implementation

The code below replicates the models and outputs reported in Section 4.

```
library(ggplot2)
library(GGally)

# Read the tsv dataset...
data <- read.delim("ccpp.tsv", header = TRUE)

# View dataset dimensions, structure, and basic summary
dim(data)
str(data)
summary(data)
head(data)

# Plot correlation matrix...
pairs_plot <- ggpairs(
  data,
  lower = list(continuous = wrap("points", alpha = 0.1, size = 0.5, color = "#2c3e50")),
  diag = list(continuous = wrap("densityDiag", fill = "#3498db", alpha = 0.5)),
  upper = list(continuous = wrap("cor", size = 4.5, color = "black"))
) +
  theme_minimal(base_size = 12) +
  theme(
    axis.text.x = element_text(angle = 90, vjust = 0.5, hjust = 1)
  ) +
  labs(title = "Combined Cycle Power Plant: Pairwise Feature Matrix")

pairs_plot
ggsave("figures/pairwise_matrix.pdf", plot = pairs_plot, width = 7, height = 7, units = "in")

# Fit a simple linear model: PE ~ AT (Ambient Temperature)
lm.fit_simple <- lm(PE ~ AT, data = data)
# Summary of the model (Coefficients, t-statistics, p-values, R-squared)
summary(lm.fit_simple)
# Extract specific components
coef(lm.fit_simple)          # Coefficients (Intercept, beta_1)
confint(lm.fit_simple)       # 95% Confidence intervals for coefficients

# Fit a multiple linear model: PE ~ AT (Ambient Temperature)
lm.fit_multiple <- lm(PE ~ ., data = data)
# Summary of the model (Coefficients, t-statistics, p-values, R-squared)
summary(lm.fit_multiple)
# Extract specific components
coef(lm.fit_multiple)        # Coefficients (Intercept, beta_1)
confint(lm.fit_multiple)     # 95% Confidence intervals for coefficients
```

```

# Predictions and Confidence/Prediction intervals for new values of AT
predict(lm.fit_simple,
        data.frame(AT = c(10, 20, 30)),
        interval = "confidence")

predict(lm.fit_simple,
        data.frame(AT = c(10, 20, 30)),
        interval = "prediction")

# Plotting simple model...
p_slr <- ggplot(data, aes(x = AT, y = PE)) +
  # Semi-transparent points to handle dense scatter
  geom_point(color = "#4A5568", alpha = 0.4, size = 1.5) +
  # Linear regression fit line with 95% Confidence Interval band
  geom_smooth(
    method = "lm",
    formula = y ~ x,
    color = "red",      # Red OLS line
    fill = "lightblue", # Shaded 95% CI
    alpha = 0.2,
    linewidth = 1.1
  ) +
  labs(
    title = "Simple Linear Regression: PE vs. AT",
    subtitle = expression(paste("Model: ", widehat(PE), " = ", beta[0], " + ", beta[1], "(AT)")),
    x = "Ambient Temperature (AT) [°C]",
    y = "Electrical Power Output (PE) [MW]"
  ) +
  theme_bw(base_size = 11) +
  theme(
    plot.title = element_text(face = "bold", size = 12),
    plot.subtitle = element_text(color = "gray30", size = 10),
    panel.grid.minor = element_blank()
  )
# Display plot
print(p_slr)
# Export for LaTeX report...
ggsave("figures/simple_regression.pdf", plot = p_slr, width = 7, height = 4.5, units = "in")

# Interaction between Ambient Temperature (AT) and Exhaust Vacuum (V)
lm.fit_interaction <- lm(PE ~ AT * V, data = data)
summary(lm.fit_interaction)

# Polynomial regression: Quadratic term for Ambient Temperature (AT^2)
lm.fit_poly <- lm(PE ~ AT + I(AT^2) + V + AP + RH, data = data)
summary(lm.fit_poly)

```

References

- James, Gareth et al. (2021). *An Introduction to Statistical Learning: with Applications in R*. 2nd. Springer Texts in Statistics. Springer. DOI: [10 . 1007 / 978 - 1 - 0716 - 1418 - 1](https://doi.org/10.1007/978-1-0716-1418-1). URL: [https : // www . statlearning . com](https://www.statlearning.com).
- Kelly, Mark, Rachel Longjohn, and Kolby Nottingham (2023). *UCI Machine Learning Repository*. University of California, Irvine, Donald Bren School of Information and Computer Sciences. URL: <https://archive.ics.uci.edu>.
- Tufekci, Pinar (2014). “Prediction of full load electrical power output of a base load operated combined cycle power plant using machine learning methods”. In: *International Journal of Electrical Power & Energy Systems* 60, pp. 126–140. URL: [https : // api . semanticscholar . org / CorpusID : 111365542](https://api.semanticscholar.org/CorpusID:111365542).
- Tufekci, Pinar and Heysem Kaya (2014). *Combined Cycle Power Plant*. UCI Machine Learning Repository. URL: <https://doi.org/10.24432/C5002N>.